

Exercise 5.1

The β -function has an expansion

$$\beta(g) = \mu \frac{dg}{d\mu} = \beta_0 g^2 + \beta_1 g^3 + \beta_2 g^4 + \dots$$

In another scheme: $g' = g + Ag^2 + Bg^3 + \dots$ and

$$\beta(g') = \beta'_0 g'^2 + \beta'_1 g'^3 + \beta'_2 g'^4 + \dots$$

$$\beta(g') = \mu \frac{d(g')}{d\mu} = \frac{dg'}{dt} = \frac{dg'}{dg} \frac{dg}{dt} = \frac{dg'}{dg} \beta(g) \quad \Bigg| \quad t = \ln \mu$$

$$\Rightarrow \beta(g') = [1 + 2Ag + 3Bg^2 + \dots] \beta(g) = \beta'_0 g'^2 + \beta'_1 g'^3 + \beta'_2 g'^4 + \dots$$

$$[1 + 2Ag + 3Bg^2 + \dots] [\beta_0 g^2 + \beta_1 g^3 + \beta_2 g^4 + \dots] = \beta'_0 [g + Ag^2 + Bg^3 + \dots]^2 + \beta'_1 [g + Ag^2 + Bg^3 + \dots]^3 + \dots$$

$$\beta_0 [g^2 + 2Ag^3 + \dots] + \beta_1 [g^3 + 2Ag^4 + \dots] = \beta'_0 [g^2 + 2Ag^3 + \dots] + \beta'_1 [g^3 + 2Ag^4 + \dots] + \dots$$

only g^2 -term
in left-hand side

only g^3 -term in
right-hand side

only g^2 -term in
right-hand side

these are the only terms with
 g^3 in the right-hand side

The right-hand side contains powers of g' so there are terms like $(Ag^2)^n$ and $(Bg^3)^n$ that aren't in left-hand side

Now we see from both sides that we have

$$\beta_0 g^2 = \beta'_0 g^2 \quad \text{and} \quad g^3 (\beta_1 + 2A\beta_0) = g^3 (\beta'_1 + 2A\beta'_0)$$

$$\Rightarrow \beta_0 = \beta'_0 \quad \Rightarrow \quad g^3 (\beta_1 + 2A\beta_0) = g^3 (\beta'_1 + 2A\beta'_0)$$

$$\Rightarrow \beta_1 = \beta'_1$$

$$\Rightarrow \text{We get } \beta_0 = \beta'_0 \quad \text{and} \quad \beta_1 = \beta'_1$$

For higher order terms than β_1 , so β_2 and higher we have scheme dependent terms ($A, B, \dots$) and their powers multiplying g^4 or terms of higher order that don't cancel with terms on left-hand side for example $(Ag^2)^2$ from $g'^2 \Rightarrow$ Only β_0 and β_1 are scheme independent

Exercise 5.2

$$F(\phi) = \int d^D r \left(\frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} r \phi^2 + \frac{1}{4} u \phi^4 \right)$$

$$r(b) = b^{-2} (r + \Delta r) \quad , \quad \Delta r = 3u \int_{b\Lambda}^{\Lambda} \frac{d^D q}{(2\pi)^D} \frac{1}{q^2 + r}$$

$$u(b) = b^{D-2} (r + \Delta u) \quad , \quad \Delta u = -9u^2 \int_{b\Lambda}^{\Lambda} \frac{d^D q}{(2\pi)^D} \frac{1}{(q^2 + r)^2}$$

Let's start by calculating the integrals

$$\Delta r = 3u \int_{b\Lambda}^{\Lambda} \frac{d^D q}{(2\pi)^D} \frac{1}{q^2 + r} = \frac{3u S_D}{(2\pi)^D} \int_{b\Lambda}^{\Lambda} dq \frac{q^{D-1}}{q^2 + r} \quad \left\| \begin{array}{l} \text{Change of} \\ \text{variables:} \\ \hat{r} = r/\Lambda^2 \\ \tilde{q} = q/\Lambda \end{array} \right.$$

$$\Delta r = 3u \frac{S_D \Lambda^{D-2}}{(2\pi)^D} \int_b^1 d\tilde{q} \frac{\tilde{q}^{D-1}}{\tilde{q}^2 + \hat{r}} \quad \left\| \begin{array}{l} \text{Let's set } D=4 \text{ except for} \\ \text{the term } \Lambda^{D-2} \\ \Rightarrow S_4 = 2\pi^2 \end{array} \right.$$

$$\Delta r = 3u \frac{2\pi^2 \Lambda^{D-2}}{2^4 \pi^4} \int_b^1 d\tilde{q} \frac{\tilde{q}^3}{\tilde{q}^2 + \hat{r}}$$

$$\Delta r = 3u \frac{\Lambda^{D-2}}{8\pi^2} \int_b^1 d\tilde{q} \left(-\frac{1}{2} \frac{d}{d\tilde{q}} [\hat{r} \ln(\tilde{q}^2 + \hat{r}) - \tilde{q}^2] \right)$$

$$\begin{aligned} \frac{d}{dx} \ln(x^2 + r) &= \frac{2x}{x^2 + r} \\ r \frac{2x}{x^2 + r} - 2x &= \frac{2xr - 2x(x^2 + r)}{x^2 + r} \\ &= -2 \frac{x^3}{x^2 + r} = -\frac{1}{2} \frac{d}{dx} (r \ln(x^2 + r) - x^2) \end{aligned}$$

$$\Delta r = 3u \frac{\Lambda^{D-2}}{8\pi^2} \frac{1}{2} \left[\tilde{q}^2 - \hat{r} \ln(\tilde{q}^2 + \hat{r}) \right] \Big|_b^1 = 3u \frac{\Lambda^{D-2}}{8\pi^2} \frac{1}{2} \left[1 - \hat{r} \ln(1 + \hat{r}) - b^2 + \hat{r} \ln(b^2 + \hat{r}) \right]$$

$$\Rightarrow \Delta r = 3u \frac{\Lambda^{D-2}}{8\pi^2} \frac{1}{2} \left[2 \ln(1/b) + \hat{r} \ln \left(\frac{b^2 + \hat{r}}{1 + \hat{r}} \right) \right] \quad \left\| \begin{array}{l} \text{here we have used} \\ b^2 = \exp(2 \ln b) \approx 1 - 2 \ln(1/b) \\ \text{for } b < 1 \end{array} \right.$$

Now let's calculate Δu :

$$\begin{aligned} \Delta u &= -9u^2 \int_{b\Lambda}^{\Lambda} \frac{d^D q}{(2\pi)^D} \frac{1}{(q^2 + r)^2} = -9u^2 \frac{S_D}{(2\pi)^D} \int_{b\Lambda}^{\Lambda} dq \frac{q^{D-1}}{(q^2 + r)^2} \quad \left\| \begin{array}{l} \tilde{q} = q/\Lambda \\ \hat{r} = r/\Lambda^2 \end{array} \right. \\ &= -9u^2 \frac{S_D \Lambda^{D-4}}{(2\pi)^D} \int_b^1 d\tilde{q} \frac{\tilde{q}^{D-1}}{(\tilde{q}^2 + \hat{r})^2} \quad \left\| \begin{array}{l} \text{we get factor of} \\ \frac{\Lambda \Lambda^{D-1}}{\Lambda^4} = \Lambda^{D-4} \end{array} \right. \\ &= -9u^2 \frac{\Lambda^{-2}}{8\pi^2} \int_b^1 d\tilde{q} \frac{\tilde{q}^3}{(\tilde{q}^2 + \hat{r})^2} \quad \left\| \begin{array}{l} D=4 \text{ except for} \\ \Lambda^{D-4} \text{ where we can} \\ S_4 = 2\pi^2 \text{ with } D=4-\epsilon \Rightarrow \Lambda^{D-4} = \Lambda^{-2} \end{array} \right. \end{aligned}$$

Using calculator to calculate the integral [to save time] we get:

$$\begin{aligned} &= -9u^2 \Lambda^{-2} \frac{1}{8\pi^2} \frac{1}{2} \left[\frac{\hat{r}}{\hat{r} + \tilde{q}^2} + \ln(\hat{r} + \tilde{q}^2) \right] \Big|_b^1 \\ &= \frac{-9u^2 \Lambda^{-2}}{8\pi^2} \frac{1}{2} \left[\frac{\hat{r}}{\hat{r} + 1} + \ln(\hat{r} + 1) - \frac{\hat{r}}{\hat{r} + b^2} - \ln(\hat{r} + b^2) \right] \end{aligned}$$

$$\Rightarrow r(b) = b^{-2} (r + \Delta r) = b^{-2} \left[r + 3u \frac{\Lambda^{D-2}}{8\pi^2} \left(\ln(1/b) + \frac{\hat{r}}{2} \ln\left(\frac{b^2 + \hat{r}}{1 + \hat{r}}\right) \right) \right]$$

$$u(b) = b^{D-4} \left[u - 9u^2 \frac{\Lambda^{-\varepsilon}}{8\pi^2} \frac{1}{2} \left[\frac{\hat{r}}{\hat{r}+1} - \frac{\hat{r}}{\hat{r}+b^2} + \ln(\hat{r}+1) - \ln(\hat{r}+b^2) \right] \right]$$

let's now use $b^2 = \exp(-2\ln(b)) \approx 1 + 2\ln(1/b)$ and

$$b^{D-4} = b^{-\varepsilon} = \exp(-\varepsilon \ln(b)) \approx 1 + \varepsilon \ln(1/b) \quad || D = 4 - \varepsilon$$

$$|\hat{r}| \ll 1 \quad \text{since } \hat{r} = r/\Lambda \quad \Rightarrow \quad \frac{\hat{r}}{1+\hat{r}} = \hat{r} \exp(-\ln(1+\hat{r})) \approx \hat{r} - \hat{r} \ln(1+\hat{r})$$

This relation holds for $|\hat{r}| \ll 1$ even if $\hat{r}$ were to be negative as $\frac{1}{1+\hat{r}} \geq 1$.

Since our approximation is valid when the argument of logarithm is near 1 which is true for $|\hat{r}| \ll 1$

$$\Rightarrow \hat{r} \ln\left[\frac{1}{1+\hat{r}}\right] \approx \frac{\hat{r}}{1+\hat{r}} - \hat{r} = 1 - \hat{r} - \frac{1}{1+\hat{r}}$$

Similarly $\hat{r} \ln(b^2 + \hat{r}) = \hat{r} \exp(\ln(b^2 + \hat{r})) \approx \hat{r} (1 + \ln(b^2 + \hat{r}))$ this also holds since $b < 1$ but not $b \ll 1$ so we are close enough of 1 for logarithm to not explode.

$$\Rightarrow \hat{r} \ln(b^2 + \hat{r}) \approx \hat{r} (b^2 + \hat{r}) - \hat{r} = \hat{r} b^2 + \hat{r}^2 - \hat{r}$$

$$\hat{r} \ln(b^2 + \hat{r}) \approx \hat{r} (1 - 2\ln(1/b)) - \hat{r} = -2\hat{r} \ln(1/b)$$

we can set $\hat{r}^2 = 0$
since $|\hat{r}| \ll 1$
let's also use $b^2 \approx 1 - 2\ln(1/b)$
Alternatively $\hat{r} \ln(b^2 + \hat{r}) \approx \hat{r} \ln(b^2)$ for $b^2 \gg \hat{r}$ works too

$$\Rightarrow r(b) = (1 + 2\ln(1/b)) \left[r + 3u \frac{\Lambda^{D-2}}{8\pi^2} \left(\ln(1/b) - \hat{r} \ln(1/b) + \frac{1}{2} \left(1 - \hat{r} - \frac{1}{1+\hat{r}} \right) \right) \right]$$

Now we need to calculate $\frac{dr(b)}{d\ln(1/b)} = \frac{dr}{d\ln(1/b)}$ to linear order in $\ln(1/b)$

To = linear order in $\ln(1/b)$:

$$\frac{dr}{d\ln(1/b)} = \frac{d}{d\ln(1/b)} \left[2\ln(1/b) r + 3u \frac{\Lambda^{D-2}}{8\pi^2} \left[\ln(1/b) - \hat{r} \ln(1/b) + \ln(1/b) - \hat{r} \ln(1/b) - \frac{1}{1+\hat{r}} \ln(1/b) \right] \right]$$

$$= 2r + 3u \frac{\Lambda^{D-2}}{8\pi^2} \left(1 - \hat{r} + 1 - \hat{r} - \frac{1}{1+\hat{r}} \right) \quad \left\{ \begin{array}{l} 2 - 2\hat{r} - \frac{1}{1+\hat{r}} = \frac{1 - 2\hat{r}^2}{1+\hat{r}} \quad || \hat{r}^2 \approx 0 \\ \approx \frac{1}{1+\hat{r}} \\ \hat{r} = r/\Lambda^2 \end{array} \right.$$

$$\frac{dr}{d\ln(1/b)} = 2r + \frac{3u}{8\pi^2} \frac{\Lambda^{D-2}}{1 + r/\Lambda^2} \quad \square$$

Now let's do same for $u(b)$

$$u(b) = (1 + \varepsilon \ln(1/b)) \left[u - g u^2 \frac{\Lambda^{-\varepsilon}}{8\pi^2} \frac{1}{2} \left(\frac{\hat{r}}{\hat{r}+1} - \frac{\hat{r}}{\hat{r}+b^2} + \ln(\hat{r}+1) - \ln(\hat{r}+b^2) \right) \right]$$

Now let's approximate:

$$\frac{\hat{r}}{\hat{r}+1} \approx \hat{r}(1 - \ln(\hat{r}+1)) = \hat{r} - \hat{r} \ln(\hat{r}+1)$$

$$- \frac{\hat{r}}{\hat{r}+b^2} \approx -\hat{r}(1 - \ln(\hat{r}+b^2)) = -\hat{r} + \hat{r} \ln(\hat{r}+b^2)$$

$$\Rightarrow \frac{\hat{r}}{\hat{r}+1} - \frac{\hat{r}}{\hat{r}+b^2} + \ln(\hat{r}+1) - \ln(\hat{r}+b^2) = (1-\hat{r})\ln(\hat{r}+1) - (1-\hat{r})\ln(\hat{r}+b^2)$$

$$= -\frac{(1-\hat{r})(1+\hat{r})}{1+\hat{r}} \left[\ln(b^2+\hat{r}) - \ln(\hat{r}+1) \right] \quad \parallel (1-\hat{r})(1+\hat{r}) = 1-\hat{r}^2 \approx 1$$

Now we use the approximations that we used previously:

$$\Rightarrow \ln(b^2+\hat{r}) \approx -2 \ln(1/b) \quad \text{and} \quad -\ln(\hat{r}+1) = \frac{1}{1+\hat{r}} - 1 = -\frac{\hat{r}}{1+\hat{r}}$$

$$\Rightarrow -(\ln(b^2+\hat{r}) - \ln(\hat{r}+1)) = -(-2 \ln(1/b) - \frac{\hat{r}}{1+\hat{r}}) = 2 \ln(1/b) + \frac{\hat{r}}{1+\hat{r}}$$

$$\Rightarrow u(b) = (1 + \varepsilon \ln(1/b)) \left[u - g u^2 \frac{\Lambda^{-\varepsilon}}{8\pi^2} \frac{1}{2} \frac{1}{1+\hat{r}} \left(2 \ln(1/b) + \frac{\hat{r}}{1+\hat{r}} \right) \right]$$

To linear order in $\ln(1/b)$ we get:

$$\frac{du(b)}{d \ln(1/b)} = u \varepsilon - g u^2 \frac{\Lambda^{-\varepsilon}}{8\pi^2} \frac{1}{2} \frac{1}{1+\hat{r}} \left(2 + \frac{\varepsilon \hat{r}}{1+\hat{r}} \right)$$

$$= u \varepsilon - g u^2 \frac{\Lambda^{-\varepsilon}}{8\pi^2} \frac{1}{2} \frac{1}{1+\hat{r}} \left[\frac{2\hat{r} + \varepsilon \hat{r} + 2}{1+\hat{r}} \right]$$

$$= u \varepsilon - g u^2 \frac{\Lambda^{-\varepsilon}}{8\pi^2} \frac{1 + \hat{r}(1 + \varepsilon/2)}{(1+\hat{r})^2}$$

$$\frac{du}{d \ln(1/b)} = u \left(\varepsilon - \frac{g u}{8\pi^2} \frac{\Lambda^{-\varepsilon}}{(1 + r/\Lambda^2)^2} \right) \quad \square$$

We can further approximate

$$\frac{1 + \hat{r}(1 + \varepsilon/2)}{(1+\hat{r})^2} \approx \frac{1}{(1+\hat{r})^2} \quad \text{since } \hat{r} \ll 1$$

$$\hat{r} = r/\Lambda^2$$

$\varepsilon \ll 1$

Exercise 5.3

The fixed point is at $\left. \frac{dr}{d \ln(1/b)} \right|_{r=r_*} = 0 = \left. \frac{du}{d \ln(1/b)} \right|_{u=u_*}$

writing $\ln(1/b) = s$ we have

$$\frac{dr}{ds} = 2r + \frac{3u}{8\pi^2} \frac{\Lambda^{D-2}}{1+\hat{r}} \quad \parallel \quad \text{let's use } \Lambda^{\epsilon} u = \hat{u} \quad \text{and } \hat{r} = r/\Lambda^2$$

$$\frac{d\hat{r}}{ds} = 2\hat{r} + \frac{3\hat{u}}{8\pi^2} \frac{\Lambda^{D-2} \Lambda^2 \Lambda^{+\epsilon}}{1+\hat{r}} \quad \parallel \quad D=4-\epsilon \Rightarrow \Lambda^{D-2} \Lambda^2 \Lambda^{+\epsilon} = \Lambda^{D-4+\epsilon} = 1$$

$$\frac{d\hat{r}}{ds} = 2\hat{r} + \frac{3\hat{u}}{8\pi^2} \frac{1}{1+\hat{r}} = 0 \Rightarrow \frac{2(1+\hat{r})\hat{r}}{1+\hat{r}} = -\frac{3}{8\pi^2} \hat{u}$$

$\hat{r} = \hat{u} = 0$ is the Gaussian fixed point

Now let's calculate $\frac{d\hat{u}}{ds}$

$$\Rightarrow \frac{d\hat{u}}{ds} = \hat{u} \left(\epsilon - \frac{9\hat{u}}{8\pi^2} \frac{1}{(1+\hat{r})^2} \right) = 0$$

$$\Rightarrow \hat{u} = 0 \quad \text{or} \quad \epsilon - \frac{9\hat{u}}{8\pi^2} \frac{1}{(1+\hat{r})^2} = 0$$

the Gaussian fixed point

$$\epsilon(1+\hat{r})^2 = \left(9/8\pi^2\right) \hat{u}$$

So now we equation pair

$$\begin{cases} \epsilon(1+\hat{r})^2 = \frac{9}{8\pi^2} \hat{u} \\ 2(1+\hat{r})\hat{r} = -\frac{3}{8\pi^2} \hat{u} \end{cases}$$

Summing both equations with also multiplying bottom one gives us following equation for $\hat{r}$:

$$\Rightarrow \epsilon(1+\hat{r})^2 + 6(1+\hat{r})\hat{r} = 0 \Leftrightarrow (1+\hat{r}) \left[\epsilon(1+\hat{r}) + 6\hat{r} \right] = 0$$

$\hat{r} = -1$ is not a valid solution for us since it produces singularities (also gives $u=0$)
So let's look int. $\epsilon(1+\hat{r}) + 6\hat{r} = 0 \Rightarrow \hat{r}(6+\epsilon) = -1$

$$\Rightarrow \hat{r} = -\frac{1}{(6+\epsilon)} \quad \text{and} \quad \hat{u} = 2 \left(1 - \frac{1}{6+\epsilon} \right) \frac{1}{6+\epsilon} \left(\frac{8\pi^2}{3} \right)$$

$$\hat{r} = -\frac{1}{(6+\epsilon)} \quad \hat{u} = \frac{2}{3} \frac{8\pi^2}{6+\epsilon} \left(1 - \frac{1}{6+\epsilon} \right)$$

$\Rightarrow$ The Wilson-Fisher fixed point is $(\hat{r}^*, \hat{u}^*) = \frac{1}{6+\epsilon} \left(-1, \frac{2}{3} 8\pi^2 \left(1 - \frac{1}{6+\epsilon} \right) \right)$

To find the flow near $(\hat{r}, \hat{u}) = (\hat{r}^*, \hat{u}^*)$ we write

$$\hat{r} = \hat{r}^* + \delta \hat{r} \quad \text{and} \quad \hat{u} = \hat{u}^* + \delta \hat{u}$$

We get the matrix T for small transformation $\delta \hat{r}$ and $\delta \hat{u}$ via

$$T \begin{pmatrix} \delta \hat{r} \\ \delta \hat{u} \end{pmatrix} = \frac{d}{ds} \begin{pmatrix} \delta \hat{r} \\ \delta \hat{u} \end{pmatrix} \quad || \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \delta \hat{r} \\ \delta \hat{u} \end{pmatrix} = \begin{pmatrix} \frac{d}{ds} \delta \hat{r} \\ \frac{d}{ds} \delta \hat{u} \end{pmatrix}$$

$$\Rightarrow a \delta \hat{r} + b \delta \hat{u} = 2 \delta \hat{r} + \frac{A \delta \hat{u}}{1 + \delta \hat{r}} \quad , \text{ where } A = \frac{3}{8\pi^2}$$

$$\Rightarrow a = 2, \quad b = \frac{A \delta \hat{u}}{1 + \delta \hat{r}} \quad \text{and}$$

$$c \delta \hat{r} + d \delta \hat{u} = \varepsilon \delta \hat{u} - \frac{B (\delta \hat{u})^2}{(1 + \delta \hat{r})^2} \quad , \text{ where } B = \frac{9}{8\pi^2}$$

$$\Rightarrow c = 0, \quad d = \varepsilon - \frac{B \delta \hat{u}}{(1 + \delta \hat{r})^2}$$

$$\Rightarrow T = \begin{pmatrix} 2 & \frac{A \delta \hat{u}}{1 + \delta \hat{r}} \\ 0 & \varepsilon - \frac{B \delta \hat{u}}{(1 + \delta \hat{r})^2} \end{pmatrix} \quad , \quad \text{We get eigenvalue } \lambda \text{ via eigenvalue equation: } \det(T - \lambda \mathbb{I}) = 0$$

$$\Rightarrow \det(T - \lambda \mathbb{I}) = \begin{vmatrix} 2 - \lambda & \frac{A \delta \hat{u}}{1 + \delta \hat{r}} \\ 0 & \varepsilon - \frac{B \delta \hat{u}}{(1 + \delta \hat{r})^2} - \lambda \end{vmatrix} = (2 - \lambda) \left(\varepsilon - \frac{B \delta \hat{u}}{(1 + \delta \hat{r})^2} - \lambda \right) = 0$$

$$\text{The two eigenvalues are } \lambda = 2 \quad \text{and} \quad \lambda = \varepsilon - \frac{B \delta \hat{u}}{(1 + \delta \hat{r})^2}$$

We get the eigenvectors v via $Tv = \lambda v$, for $\lambda = 2$ we get

$$\Rightarrow \begin{pmatrix} 2 & \frac{A \delta \hat{u}}{1 + \delta \hat{r}} \\ 0 & \varepsilon - \frac{B \delta \hat{u}}{(1 + \delta \hat{r})^2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix} \quad \left| \begin{array}{l} \text{Let's write } A' = A \frac{\delta \hat{u}}{1 + \delta \hat{r}} \\ B' = \varepsilon - \frac{B \delta \hat{u}}{(1 + \delta \hat{r})^2} \end{array} \right.$$

$$\Rightarrow \begin{cases} 2x + A'y = 2x \\ B'y = 2y \end{cases} \Rightarrow \begin{cases} x = x \\ y = 0 \end{cases} \quad \begin{array}{l} \text{let's set } x = 1 \\ \text{The first eigenvector} \\ \text{is} \end{array} \quad v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

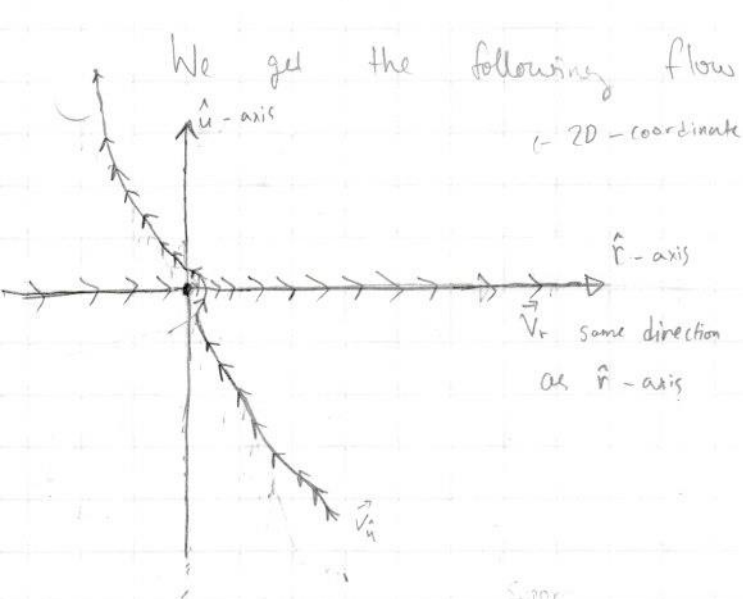
Now let's find v_2 for $\lambda = B'$, $T v_2 = B' v_2$

$$\Rightarrow \begin{cases} 2x + A'y = B'x \\ B'y = B'y \end{cases} \Rightarrow \begin{cases} x = \frac{A'}{2+B'} \\ y = y \end{cases}, \text{ let's set } y=1$$

$$\Rightarrow v_2 = \left(\frac{A'}{2+B'}, 1 \right)$$

So the eigenvalues are $\lambda_r = 2$ and $\lambda_u = \varepsilon - \frac{B \delta \hat{u}}{(1+\delta \hat{r})^2}$

and the eigenvectors are: $v_r = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v_u = \left(A \frac{\delta \hat{u}}{1+\delta \hat{r}} \cdot \frac{1}{2 \left(\varepsilon - \frac{B \delta \hat{u}}{(1+\delta \hat{r})^2} \right)}, 1 \right)$



$D > 4$ case $\Rightarrow \varepsilon < 0$

$\vec{v}_u$ always flows positively (up) in $\hat{u}$ -axis.

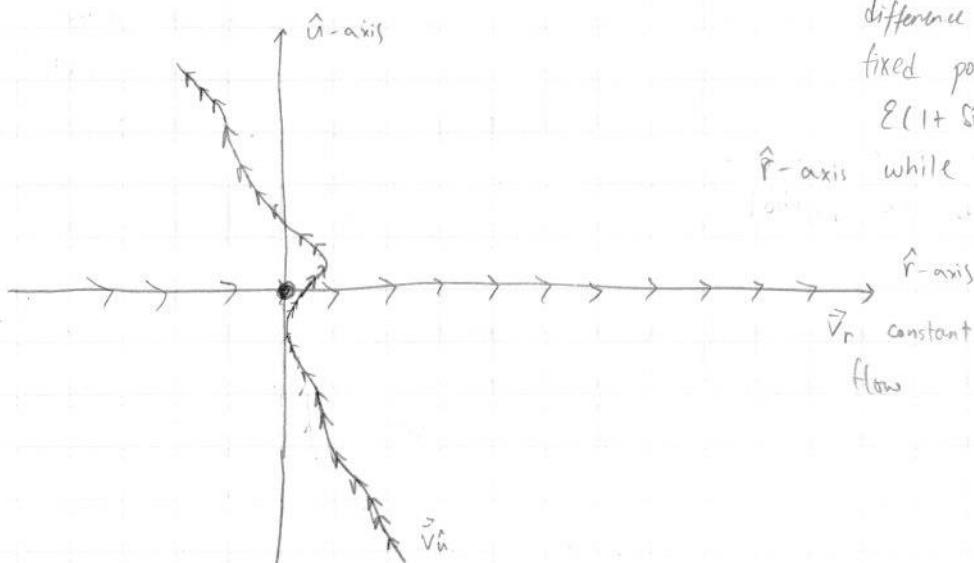
$\vec{v}_u$ flows positively in $\hat{r}$ -axis when

$$\frac{A \delta \hat{u} (1+\delta \hat{r})}{2 \left(\varepsilon (1+\delta \hat{r})^2 - B \delta \hat{u} \right)} > 0$$

"

$\vec{v}_u$ flows negatively in $\hat{r}$ -axis when $\delta \hat{u} > 0$ until $\delta \hat{r}$ reaches -1.

Now a diagram for $D < 4 \Rightarrow \varepsilon > 0$



Since $\varepsilon > 0$ there is small difference of flow near the fixed point and now the $\hat{u}$ along with $\varepsilon (1+\delta \hat{r})^2$ will suppress flow along $\hat{r}$ -axis while $\hat{u} (1+\delta \hat{r})$ try to increase it

Exercise 5.4

$$\mathcal{L}_{EH} = \mathcal{L}_0 + \mathcal{L}_{d=8}, \text{ where } \mathcal{L}_0 = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

and $\mathcal{L}_{d=8}$ is the interaction Lagrangian of dimension 8. We also have $E \ll m_e$

At low energies $E \ll m_e$ we can integrate out all charged fermion contribution.

So $\mathcal{L}_{EH} = \mathcal{L}_{EH}(A)$, only photons contribute.

(Mass-dimension of our Lagrangian must be 4 is 4-dimensions)

Because of Gauge invariance we may only use $F_{\mu\nu}$ and $\tilde{F}_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} F_{\alpha\beta}$ to construct the Lagrangian. Let's start with following effective Lagrangian:

$$\mathcal{L}_{EH} = \Lambda^2 \mathcal{L}_{d=2} + \mathcal{L}_{d=4} + \frac{1}{\Lambda^2} \mathcal{L}_{d=6} + \frac{1}{\Lambda^4} \mathcal{L}_{d=8} + \dots$$

$\left\{ \begin{array}{l} [\mathcal{L}] = +1 \\ d = \text{mass dimension} \\ \text{and } [F_{\mu\nu}] = 2 \text{ mass dimension} \\ [A_\mu] = 1, [\partial_\mu] = 1 \end{array} \right.$

$$\mathcal{L}_{d=2} = 0 \text{ since only } F_\mu{}^\mu \text{ or } \tilde{F}_\mu{}^\mu \text{ is available and } F_\mu{}^\mu = \partial_\mu A^\mu - \partial^\mu A_\mu = 0$$

$$\text{and } \tilde{F}_\mu{}^\mu = \frac{1}{2} \epsilon_{\mu}{}^{\alpha\beta} F_{\alpha\beta} = 0$$

$$\mathcal{L}_{d=4} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \text{ here } F_{\mu\nu} \tilde{F}^{\mu\nu} \text{ can be discarded since it is a total derivative. (total derivatives satisfy Euler-Lagrange equation)}$$

$$\text{Another possible term is } \tilde{F}_{\mu\nu} \tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} F_{\alpha\beta} \frac{1}{2} \epsilon^{\mu\nu\gamma\delta} F_{\gamma\delta} = \frac{1}{4} (\epsilon_{\mu\nu\alpha\beta} \epsilon^{\mu\nu\gamma\delta}) F_{\alpha\beta} F_{\gamma\delta}$$

$$\epsilon_{\mu\nu\alpha\beta} \epsilon^{\mu\nu\gamma\delta} = \delta_{\alpha\gamma} \delta_{\beta\delta} - \delta_{\alpha\delta} \delta_{\beta\gamma} \text{ for } \alpha \neq \beta \text{ and } 0 \text{ for } \alpha = \beta$$

$$\Rightarrow \tilde{F}_{\mu\nu} \tilde{F}^{\mu\nu} = \frac{1}{4} (F_{\alpha\beta} F_{\alpha\beta} + F_{\alpha\beta} F_{\beta\alpha}) = 0 \text{ since } F_{\alpha\beta} = -F_{\beta\alpha} \text{ (also } F_{\alpha\alpha} = 0 \text{ always)}$$

$$\Rightarrow \mathcal{L}_{d=4} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \mathcal{L}_0$$

Now let's see $\mathcal{L}_{d=6}$, which has these possibilities:

$$F_{\mu\nu} F^{\nu\rho} F_{\rho\mu} = -F_{\mu\rho} F^{\rho\nu} F_{\nu\mu} = -F_{\mu\rho} F^{\rho\nu} F_{\nu\mu} \quad \parallel \text{ Since } F_{\mu\nu} = -F_{\nu\mu}$$

$$F_{\mu\nu} F^{\nu\rho} F_{\rho\mu} = -F_{\mu\rho} F^{\rho\nu} F_{\nu\mu} \leftarrow \text{this is just renaming of } \rho \leftrightarrow \nu \text{ so } F_{\mu\nu} F^{\nu\rho} F_{\rho\mu} = F_{\mu\rho} F^{\rho\nu} F_{\nu\mu}$$

$$\Rightarrow F_{\mu\rho} F^{\rho\nu} F_{\nu\mu} = -F_{\mu\rho} F^{\rho\nu} F_{\nu\mu} \Rightarrow F_{\mu\nu} F^{\nu\rho} F_{\rho\mu} = 0, \text{ all odd \# of } F_{\mu\nu} \text{ contractions vanish}$$

We also have $\tilde{F}_{\mu\nu} = \tilde{F}_{\nu\mu}$ since $\epsilon_{\mu\nu\alpha\beta} = -\epsilon_{\nu\mu\alpha\beta}$, which means

$\Rightarrow$ All odd $\tilde{F}_{\mu\nu}$ contractions vanish for same reason as for $F_{\mu\nu}$

All odd contractions also vanish when $F_{\mu\nu}$ and $\tilde{F}_{\mu\nu}$ are mixed for

example: $F_{\mu\nu} F^{\nu\rho} \tilde{F}_{\rho\mu} = 0$ etc.

Now let's show that $\mathcal{L}_{d=8}$ is non-zero

$\mathcal{L}_{d=8}$ can be built from $F_{\mu\nu} \tilde{F}^{\mu\nu}$ and $F_{\mu\nu} F^{\mu\nu}$ alone to give us linearly independent operators.

We can have $(F_{\mu\nu} F^{\mu\nu})^2 \neq 0$ since $F_{\mu\nu} F^{\mu\nu}$ is not zero and appears in our $\mathcal{L}_0$.

Similarly we can have $(F_{\mu\nu} \tilde{F}^{\mu\nu})^2 \neq 0$ now since we have power of 2 of $F_{\mu\nu} \tilde{F}^{\mu\nu}$ it is no longer a total derivatives so we can't discard it.

The third linearly independent term is $(F_{\mu\nu} F^{\mu\nu})(F_{\alpha\beta} \tilde{F}^{\alpha\beta})$

$$(F_{\mu\nu} F^{\mu\nu}) = E^2 - B^2 \quad \text{and} \quad F_{\alpha\beta} \tilde{F}^{\alpha\beta} = \vec{E} \cdot \vec{B}$$

$$\Rightarrow (F_{\mu\nu} F^{\mu\nu})(F_{\alpha\beta} \tilde{F}^{\alpha\beta}) = (E^2 - B^2)(\vec{E} \cdot \vec{B})$$

QED is parity conserving theory but $\vec{B}$ is odd under parity while E is even
 $\Rightarrow \vec{E} \cdot \vec{B}$ doesn't conserve parity. This is because $\vec{B}$ is pseudo vector while $\vec{E}$ is a vector. Since $(F_{\mu\nu} F^{\mu\nu})(F_{\alpha\beta} \tilde{F}^{\alpha\beta})$ doesn't conserve parity it is forbidden so we only have two linearly independent terms that make up $\mathcal{L}_{d=8}$

$$\Rightarrow \mathcal{L}_{d=8} = g_1 (F_{\mu\nu} F^{\mu\nu})^2 + g_2 (F_{\mu\nu} \tilde{F}^{\mu\nu})^2, \quad \text{with } g_1 \text{ and } g_2 \text{ being effective couplings}$$

$$\text{So we had } \mathcal{L}_{\text{Ef}} = \Lambda^2 \underbrace{\mathcal{L}_{d=2}}_{=0} + \mathcal{L}_{d=4} + \underbrace{\frac{1}{\Lambda^2} \mathcal{L}_{d=6}}_{=0} + \frac{1}{\Lambda^4} \mathcal{L}_{d=8}$$

$$\mathcal{L}_{\text{Ef}} = \mathcal{L}_{d=4} + \frac{1}{\Lambda^4} \mathcal{L}_{d=8}, \quad \text{let's absorb } \frac{1}{\Lambda^4} \text{ into } g_1 \text{ and } g_2$$

$$\Rightarrow \text{to lowest order in interactions: } \mathcal{L}_{\text{Ef}} = \mathcal{L}_0 + \mathcal{L}_{d=8} \quad \text{with}$$

$$\mathcal{L}_{d=8} = g_1 (F_{\mu\nu} F^{\mu\nu})^2 + g_2 (F_{\mu\nu} \tilde{F}^{\mu\nu})^2$$
